

Lecture 1: Getting started with supervised learning

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Machine Learning I: 46-926

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Goals of this course

- ▶ Become familiar with common machine learning tools and ideas, from both a **theoretical** foundation and an **applied** viewpoint
- ▶ Be able to recognize a problem and develop a useful approach to modeling it, and then actually carry it out
- ▶ Become comfortable with the fundamentals of machine learning, so that more complicated approaches can be understood and incorporated in the future

This course: practical aspects

- ▶ 6 homeworks, due on Canvas at the beginning of classes on Wednesdays starting week 2.
 - ▶ I really mean the **beginning**, since I want to discuss the homeworks at the start of class.
 - ▶ The homeworks will tend to require a significant effort...
- ▶ Short weekly quizzes on the material on Fridays, starting next week. These will take the place of a final exam this mini for grading, and can be taken asynchronously on Canvas.
 - ▶ This is new and is intended to make life better, so feedback is welcome.
- ▶ Notes will be posted before class so you can write on them. Scanned/tablet versions of the notes will be posted after class (by next day).
 - ▶ Will often post more than we will cover, just in case.

Homeworks

Homework problems take three main forms:

- ▶ **Theoretical problems** to understand how things work and to become familiar with important concepts and tools. Gives insight into when they should do well or fail, and guides the construction of future methods.
- ▶ **Simulations** to see how methods perform – and more often, how they break down.
- ▶ **Data examples** to see how to run methods and to build an intuition about what you would see on real data. Also helps to understand problems that arise in practical settings and how to fix them.

Collaboration

Talk about the homeworks, help each other learn. However, you need to:

- ▶ Write your own code. Making debugging suggestions is ok. Writing one version of the code together is not.
- ▶ Do your own writeup of the math. You can talk about and sketch ideas, but you shouldn't be looking at someone else's work when you are writing your own.

The quizzes must be your own work.

In both cases, you should be able to explain anything that you submit. . .

Discussion

There is a Piazza discussion board through canvas. Please use this instead of email for any questions about the material or homework.

This way there is a fair distribution of information, and you can also help each other learn.

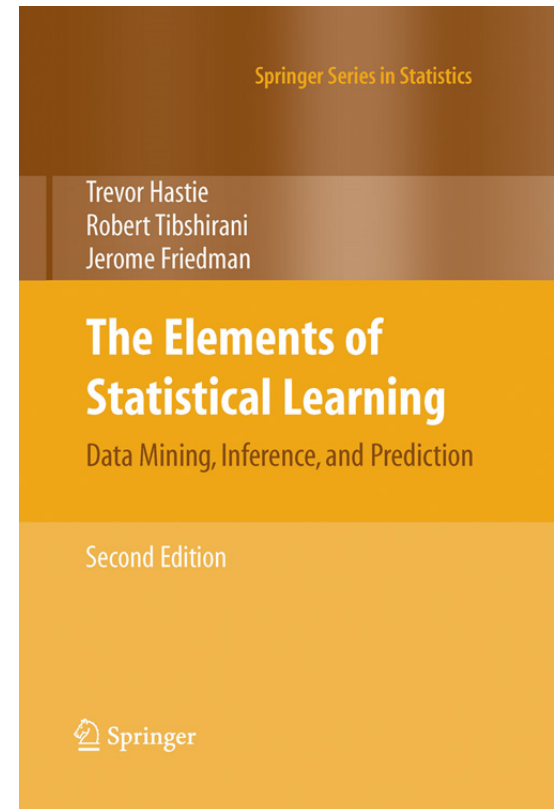
I also tend to be faster at Piazza than at email. However, I make no promises with either very close to homework deadlines.

Books

Two useful references for this course are

- ▶ *Elements of Statistical Learning* by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
- ▶ *Introduction to Statistical Learning* by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. It is easier to read and has more code examples, but covers less material and is written at a lower level.

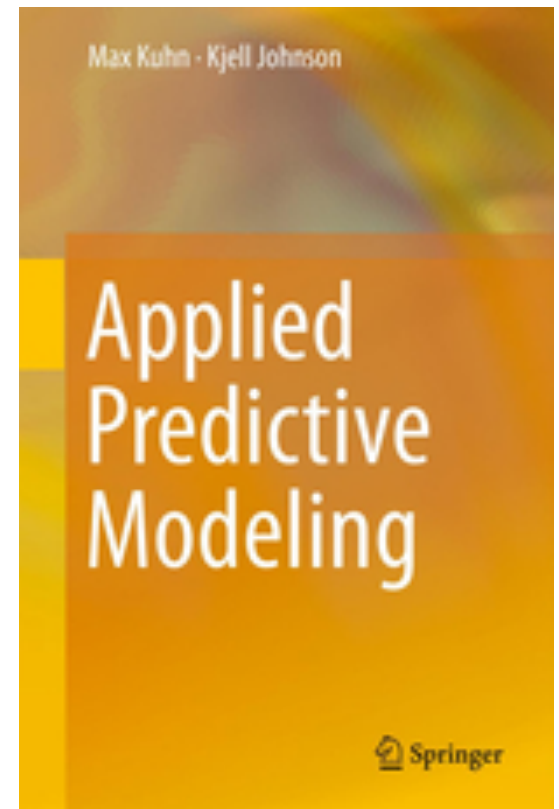
Both textbooks are available (legally) for free online from the authors. Both are also available from SpringerLink if you are on the CMU network.



Books

The book *Applied Predictive Modeling* by Max Kuhn and Kjell Johnson is also a useful reference for some of the more practical aspects of applying machine learning.

This book is available free online from SpringerLink if you are on the CMU network.



Subject matter

- ▶ While this is an machine learning class with a focus on finance, our primary goal will be the machine learning. If there is a chance to better illustrate an interesting concept with a non-finance data set, I will do so.
 - ▶ Many methods were originally developed for other uses, which can lend a better intuition.
 - ▶ Other fields often have higher signal-to-noise, making for a cleaner illustration of new ideas.
 - ▶ Interesting new applications in finance take time to spread.
- ▶ The course is planned to be done primarily in python. However, for some methods R is better suited, so we will fall back on that from time to time as an alternative or just for comparison.

Announcements

- ▶ There will be an announcement slide at the beginning of each lecture. You are responsible for being aware of this content, even if you miss class.
- ▶ Homework 1 is out Wednesday and due on Wednesday.



Today: Intro to ML and Supervised Learning

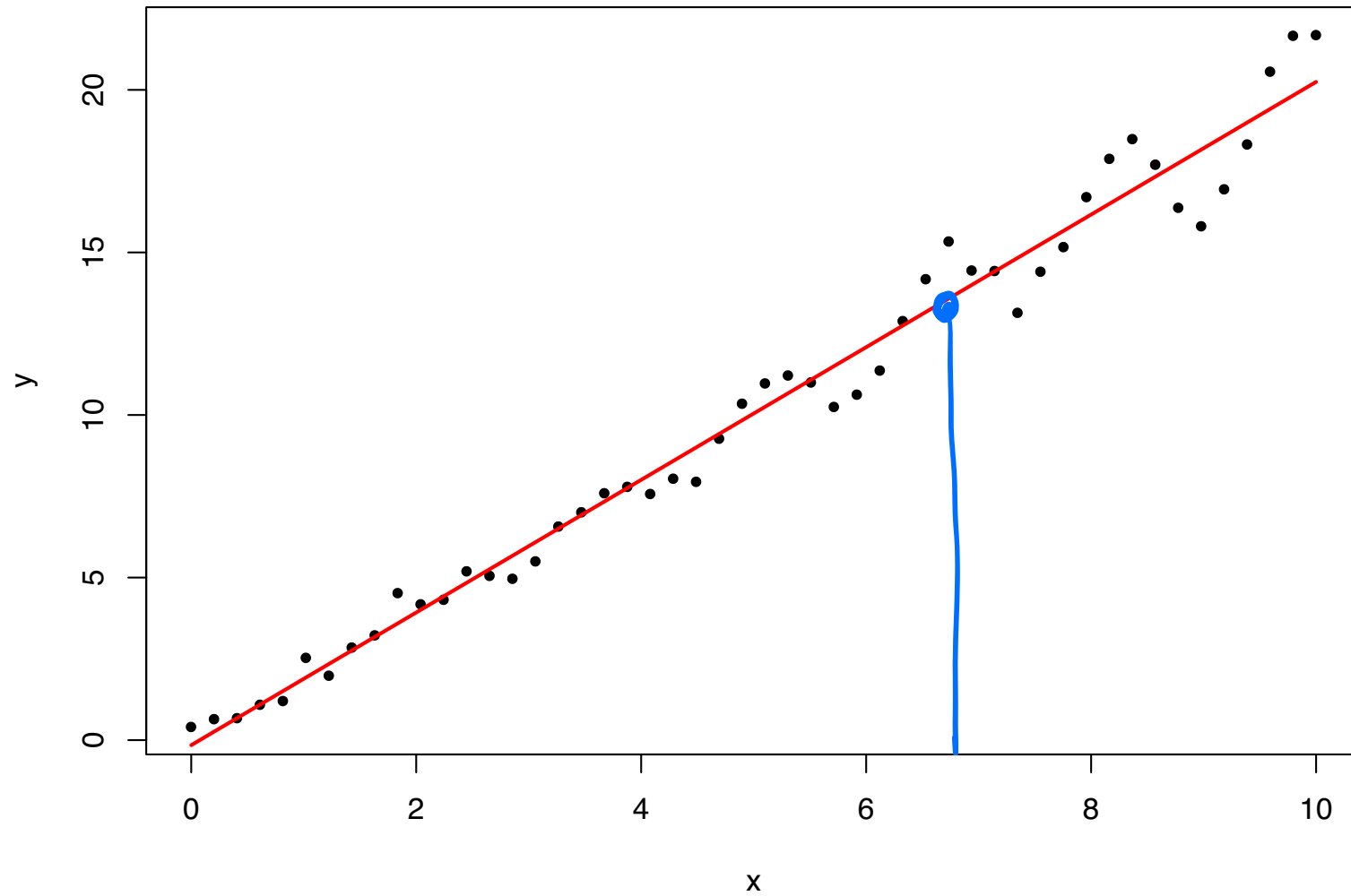
- ▶ The basic terminology and intuitions behind Machine Learning
- ▶ Returning to linear regression
- ▶ Extending linear regression

Classification of problems

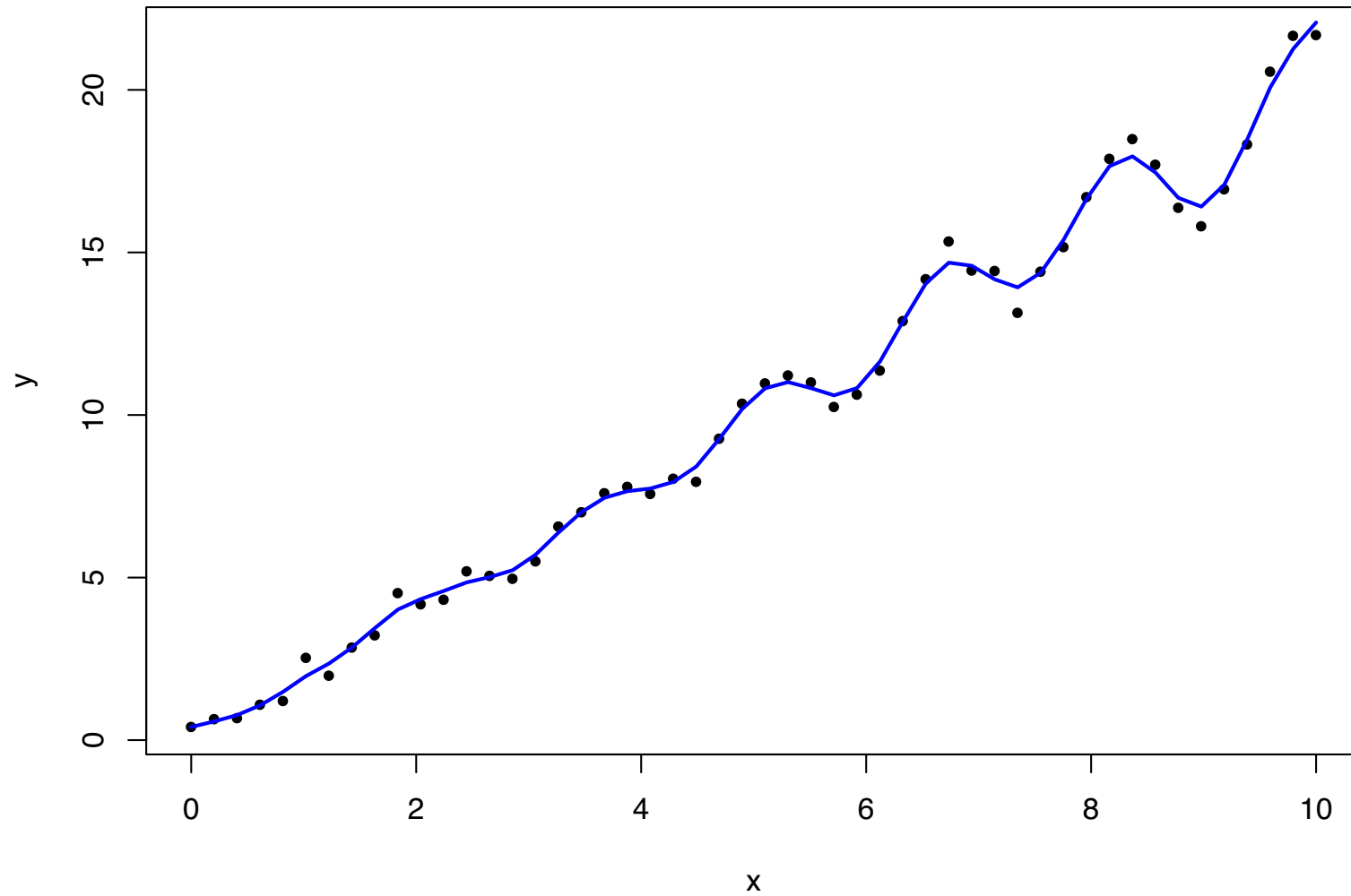
Machine learning problems are often (over simplistically) divided as follows

- ▶ **Supervised learning**: Making predictions
i.e., given measurements $(X_1, Y_1), \dots, (X_n, Y_n)$, learn a model to predict Y_i from X_i
 - ▶ **Prediction**: Y_i is a continuous value
 - ▶ **Classification**: Y_i is a discrete value
- ▶ **Unsupervised learning**: discovering structure
E.g., given measurements $X_1, \dots, X_n$, learn some underlying structure of the X distribution.

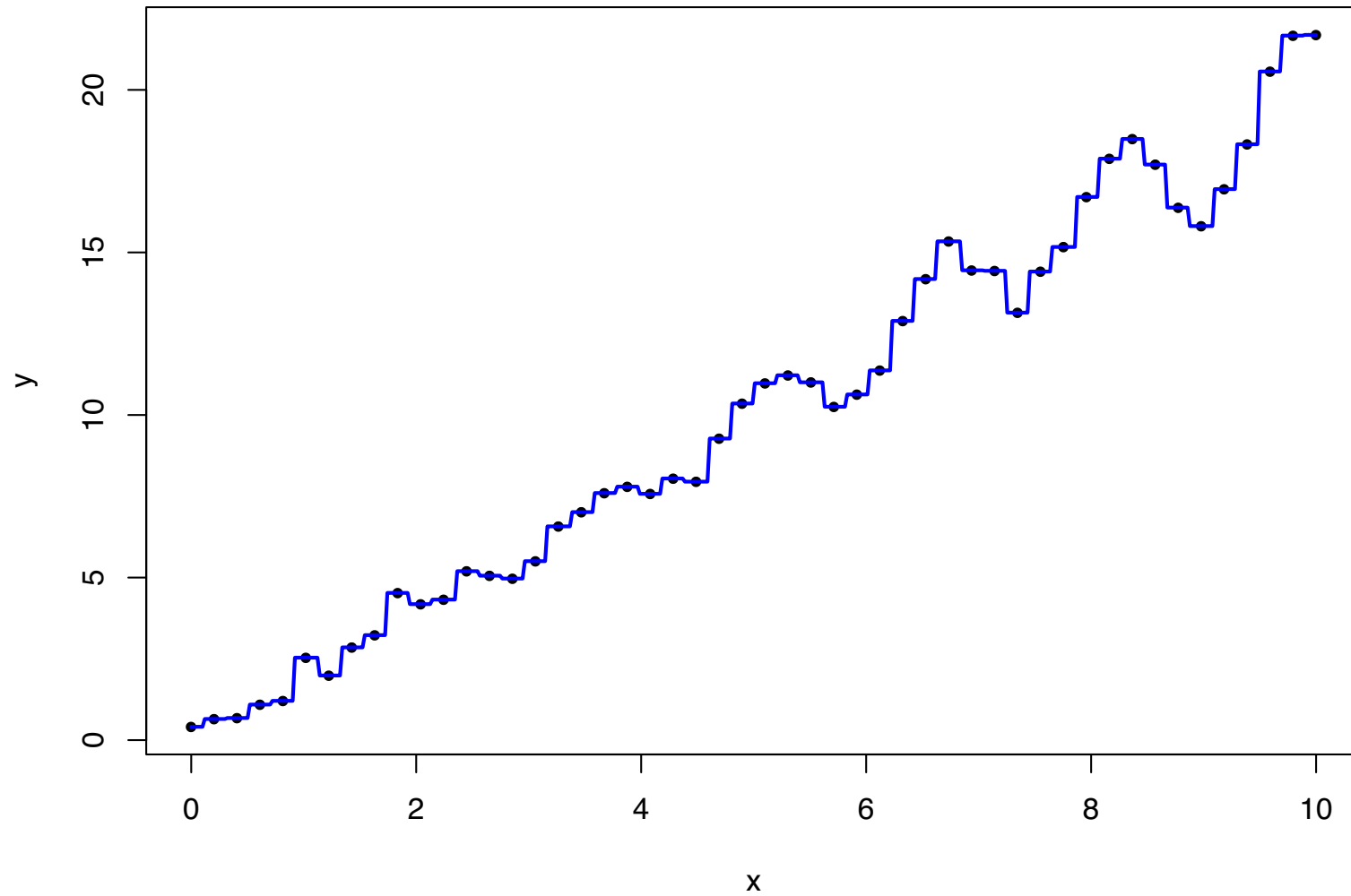
Supervised learning: Linear regression



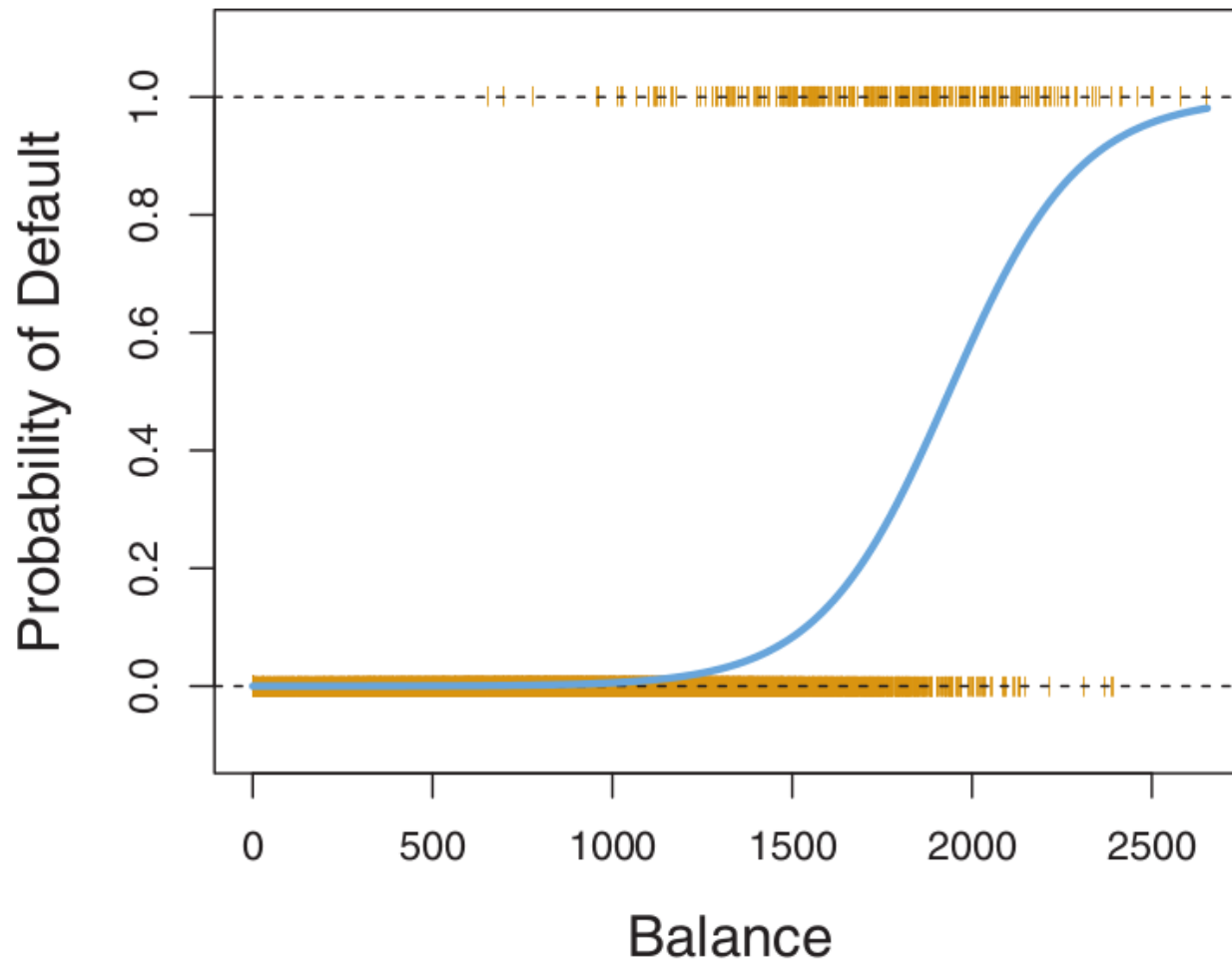
Supervised learning: Splines



Supervised learning: k -nearest neighbors

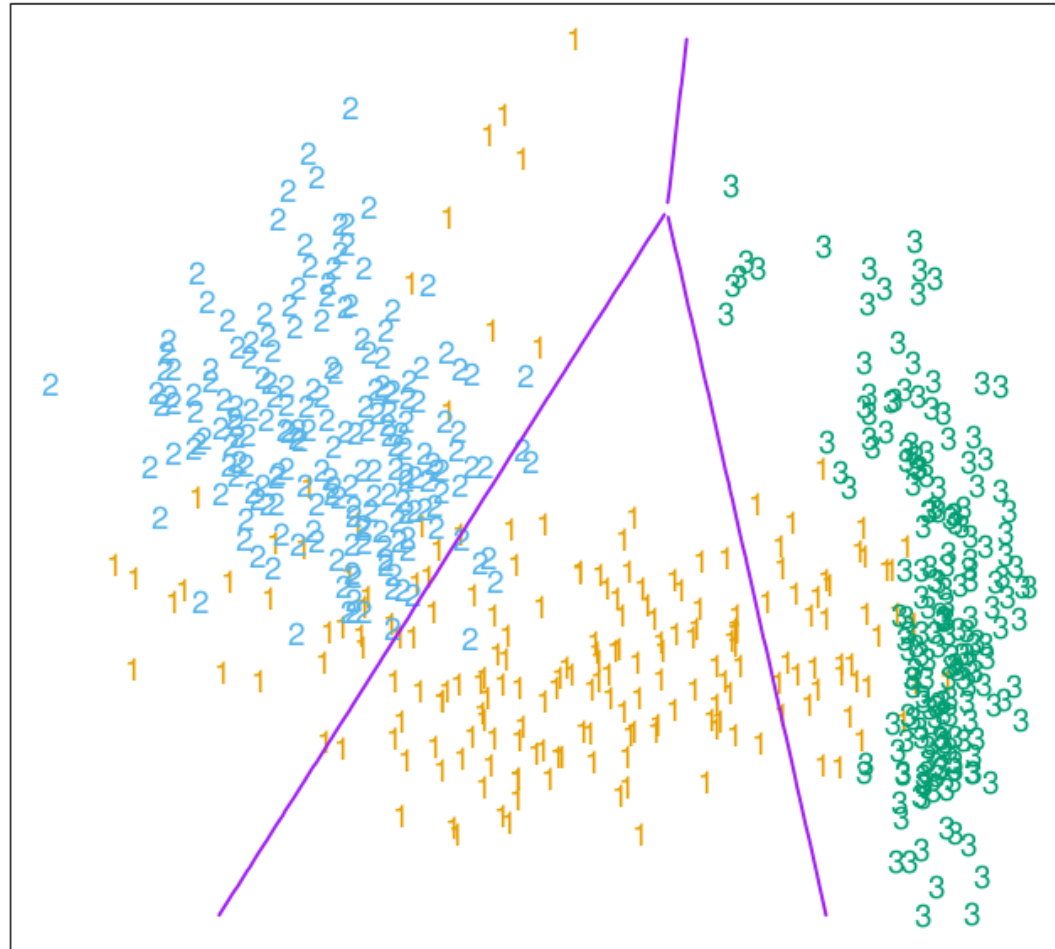


Supervised learning: Logistic regression



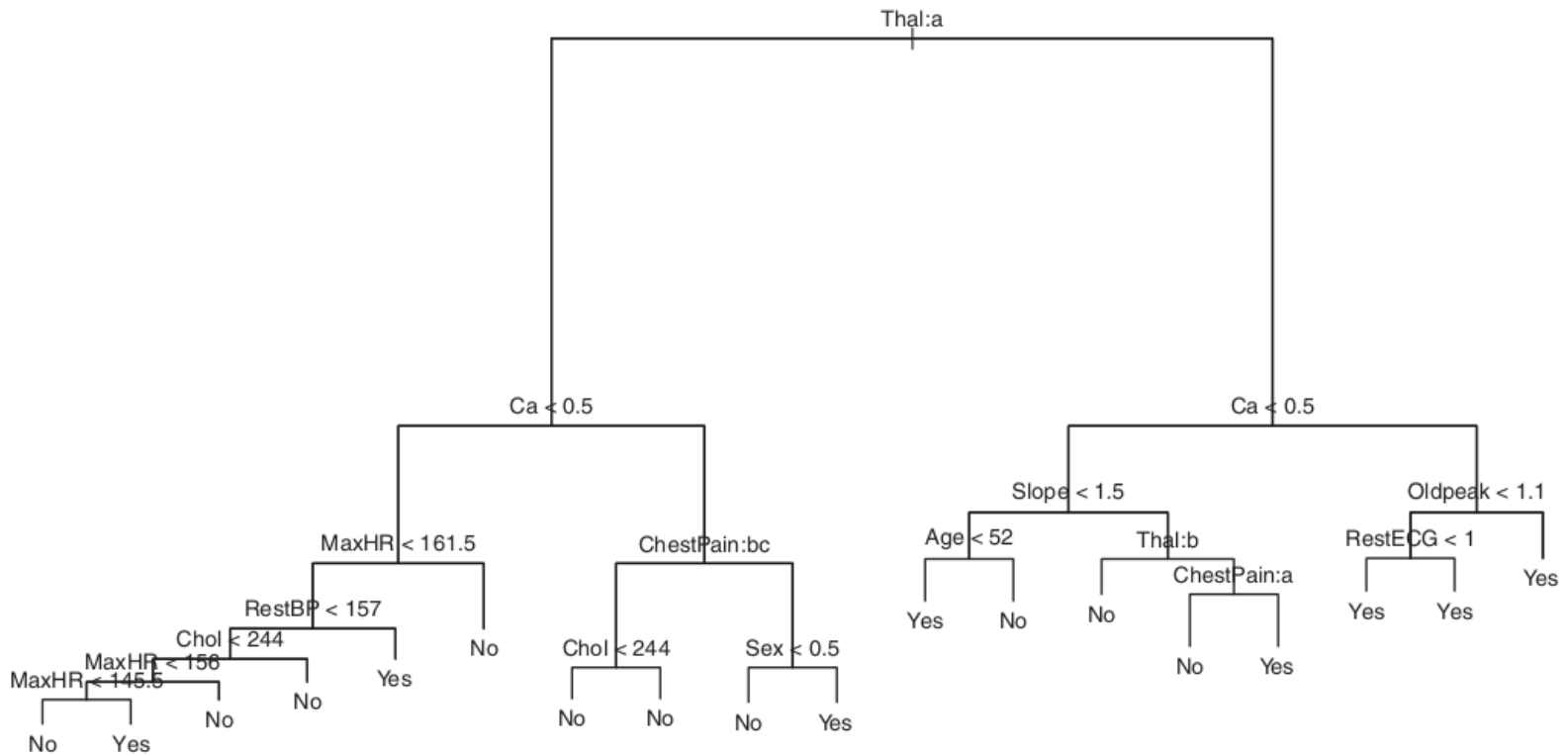
(ISL pg. 131)

Supervised learning: Linear discriminant analysis



(ESL pg. 103)

Supervised learning: Decision trees, forests, boosting



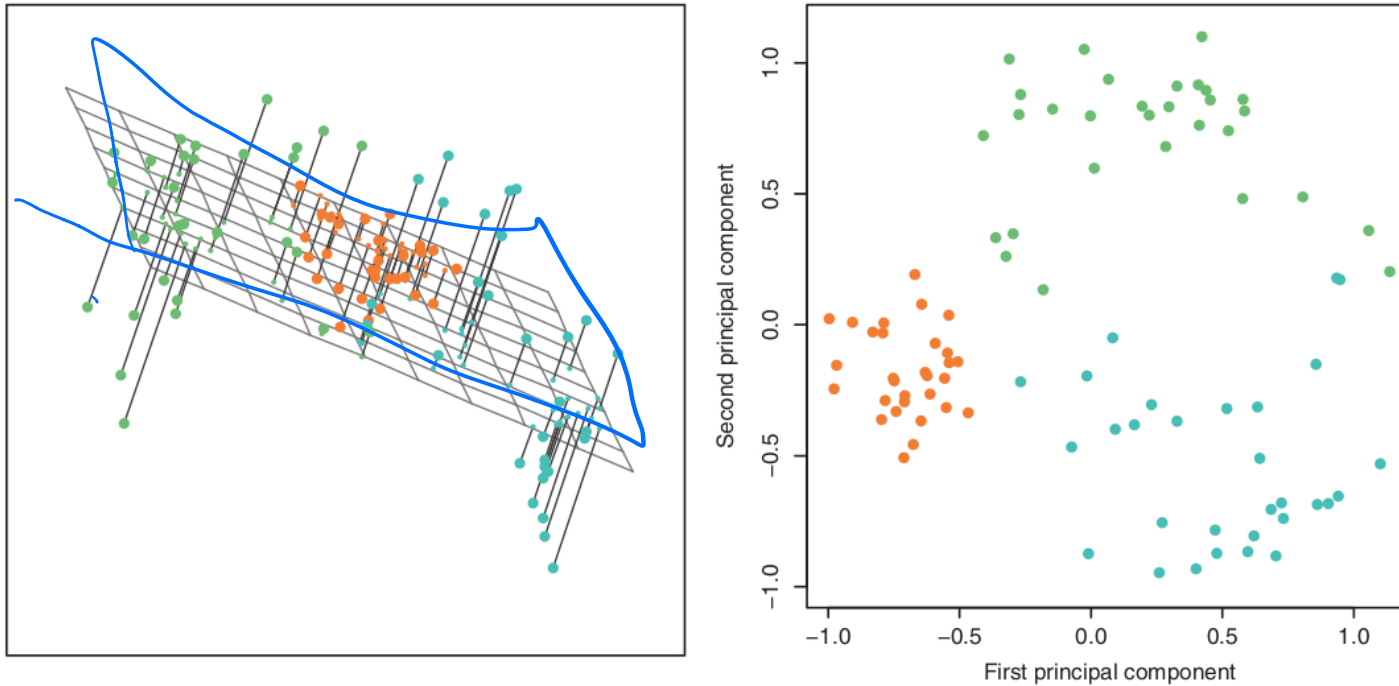
(ISL pg. 313)

Unsupervised learning

Unsupervised learning sounds a little more vague. What does it mean to “discover underlying structure”?

- ▶ **Dimension reduction** Discover lower-dimensional structure in high dimensional data
- ▶ **Clustering** Detect groups of similar units or variables

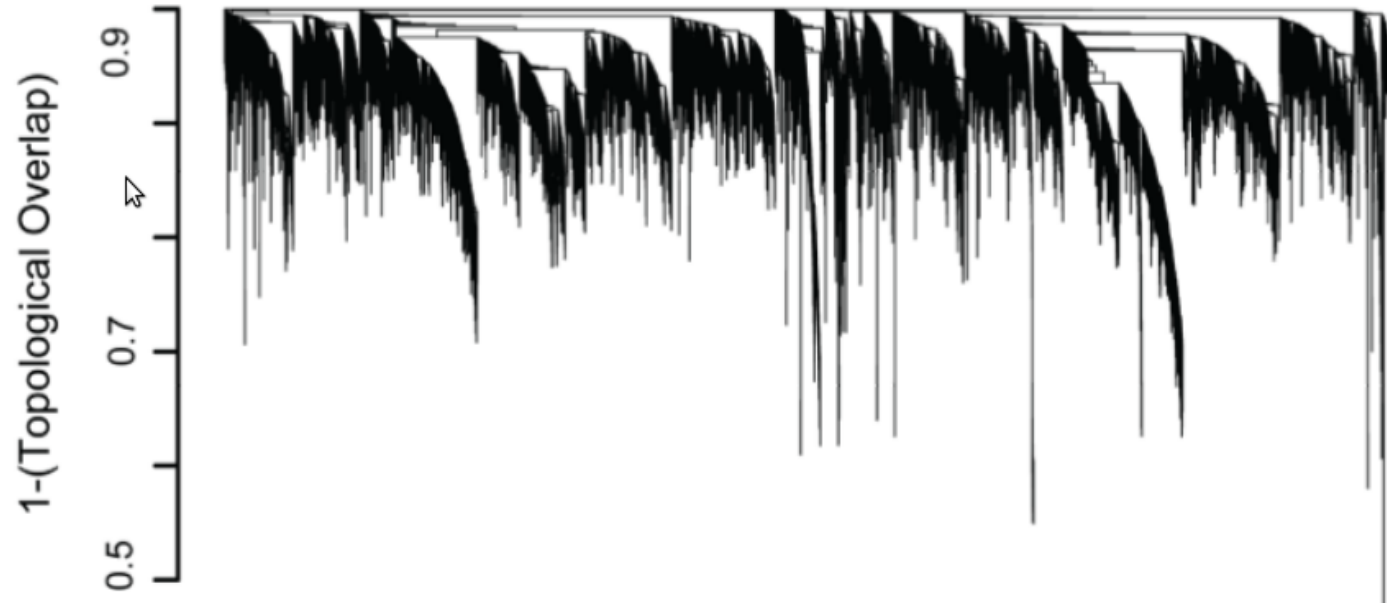
Unsupervised learning: PCA



(ISL pg. 131)

Discover lower-dimensional structure in high dimensional data

Unsupervised learning: Clustering



(Parikshak et. al., 2013)

Detect groups of similar units or variables

Supervised/unsupervised learning

These divisions are somewhat limiting. There is plenty of crossover and ambiguity.

- ▶ Is search a supervised or unsupervised problem?
- ▶ If I detect low dimensional structure in the covariates, shouldn't I be able to make better predictions?
- ▶ Reinforcement learning learns a policy in an online fashion, sampling input/output pairs to both:
 - ▶ Learn about the system
 - ▶ Maximize payoff, including these samples

Nevertheless, the terms provide a useful approach to thinking about problems and putting them in a reasonable framework. We'll start there, and move on to stranger topics (mostly ML2).

Supervised learning setup

In the basic prediction/classification ML setup, we observe independent draws (x, y) from a fixed distribution $\mathcal{P}$, with $x \in \mathbb{R}^p$.

- ▶ In *prediction*, $y \in \mathbb{R}$
- ▶ In *classification*, $y \in \{1, \dots, K\}$, or some other discrete set.

In the simplest setting, we see n examples, $(x_1, y_1), \dots, (x_n, y_n)$. From these, we want to build a function $\hat{\mu}(x)$ that takes vectors $X \in \mathbb{R}^p$ to guesses at Y .

- ▶ X as something we can get, either because it happened in the past or because it's cheap to measure.
- ▶ Y is something we want but is hard to get, either because it will happen in the future or is expensive to measure.

In later lectures, we will see interesting modifications of this setup, and the resulting complications.

A few examples

Many problems appear in finance that fit in a prediction framework. You might:

- ▶ Predict future company fundamentals
- ▶ Predict whether a mortgage ends in Default^P/Repayment/Sale
- ▶ Estimate how long a mortgage will remain active before such an event
- ▶ Classify fraudulent or suspicious behavior
- ▶ Estimate cost of execution or select execution method
- ▶ Predict future customer value
- ▶ Estimate probability of customer acquisition

It's useful to look for settings where there is likely to actually be signal. E.g., predicting price movements after an interesting event, rather than just daily price movements.

What makes a good $\hat{\mu}$?

For a given (X, Y) pair drawn from $\mathcal{P}$, we observe X and use $\hat{\mu}(X)$ to estimate Y .

It is unlikely that our guesses $\hat{\mu}(X)$ will always be perfect. How should we think about their quality?

- ▶ Is it equally bad to classify a valid transaction as fraud as it is to classify a fraudulent transaction as valid?
- ▶ How much worse is it to miss your estimate of execution cost by a factor of two? Twice as bad?
- ▶ Is it equally bad to overestimate future earnings and to underestimate future earnings?

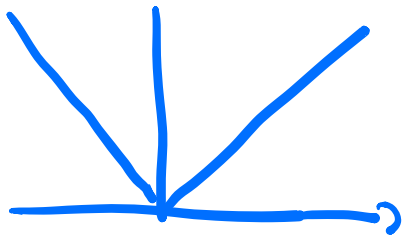
We use the Loss function to encode these tradeoffs.

Loss function

The *loss function*, $\mathcal{L}(\underline{Y}, \underline{\hat{\mu}(X)})$, represents the cost of guessing $\hat{\mu}(X)$ when the true value is Y (for a new (X, Y) from $\mathcal{P}$).

Examples:

- Absolute deviation

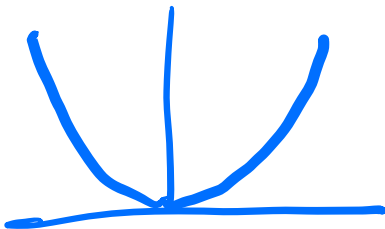


$$\mathcal{L}(Y, \hat{\mu}(X)) = |Y - \hat{\mu}(X)|$$

$$\frac{|Y - \hat{\mu}(X)|}{Y}$$

relative loss

- Squared error



$$\mathcal{L}(Y, \hat{\mu}(X)) = (Y - \hat{\mu}(X))^2$$

- 0-1 Loss (classification)

$$\mathcal{L}(Y, \hat{\mu}(X)) = \mathbb{1}_{Y \neq \hat{\mu}(X)}$$

Loss function

The *loss function*, $\mathcal{L}(Y, \hat{\mu}(X))$, represents the cost of guessing $\hat{\mu}(X)$ when the true value is Y (for a new (X, Y) from $\mathcal{P}$).

For convenience, we often use *squared error loss* for prediction,

$$\mathcal{L}(Y, \hat{\mu}(X)) = (Y - \hat{\mu}(X))^2$$

and 0 - 1 loss for *classification*,

$$\mathcal{L}(Y, \hat{\mu}(X)) = \mathbf{1}_{Y \neq \hat{\mu}(X)}$$

Note that these are *rarely your actual losses*. Nevertheless, they are useful for thinking about and developing methods. You should always stop and think about the implications.

Realistic Loss

Throughout this class (and machine learning as a whole), we make many idealizations, including our loss functions.

This is fine, as long as you understand this limitation and keep it in mind. You should always think about what your actual loss, and at least stop at the end to consider the consequence of ignoring it.

- ▶ How might this incorrect loss drive my estimates in an undesirable direction?
- ▶ Compare the final performance for a more realistic loss. You can compute test error or cross-validation error (later) on losses that are different from your training loss.

Expected Loss

$$= \frac{(y - \hat{\mu}(x))^2}{n}$$

Of course, it's going to be hard to assess $\mathcal{L}(Y, \hat{\mu}(X))$ directly, since it's actually a random variable! How should I optimize that?

Instead, we tend to focus on a summary measure that's easy to estimate. We tend to use expected loss, $\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X))$.

I could consider the expected loss for a fixed function μ , in which case our expectation is over $(X, Y) \sim \mathcal{P}$.

However, our actual $\hat{\mu}$ is usually also random! We will often consider the expected loss over $\hat{\mu}, X, Y$.

- ▶ Imagine sampling a random data set, X, Y .
- ▶ Fitting a $\hat{\mu}$ to that data.
- ▶ Drawing a point X, Y to evaluate the function $\hat{\mu}$ at.

Expected Loss

Expected loss is the most common metric one tries to optimize:
Expected prediction error (regression):

$$\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X)) = E[(Y - \hat{\mu}(X))^2]$$

Expected misclassification error (classification):

$$\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X)) = E[\mathbb{1}_{Y \neq \hat{\mu}(X)}]$$

Even though these may not completely reflect our true cost, they give us a very useful way to think about machine learning. In particular, they lead to two simple, important insights.

Regression function (x, y)

Suppose we had our dream, and we actually knew $\mathcal{P}$. What would be the best function μ we could possibly use? Let's minimize the expected prediction error. Here X, Y are random, but not μ !

$$\int f(\mu(x)) dx$$

$$\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X)) =$$

$$E[(Y - \mu(X))^2] = E[E[(Y - \mu(X))^2 | X]]$$

$$= E[E[Y^2 - 2\mu(X)Y + \mu(X)^2 | X]]$$

$$= E[E(Y^2 | X) - 2\mu(X)E(Y | X) + \mu(X)^2]$$

$$= \text{minimize over } \mu(x)$$

minimize $z^2 - 2cz$
 $2z - 2c = 0$
 $z = c$

The first term doesn't depend on μ . Minimizing the other two with respect to $\mu(X)$ gives

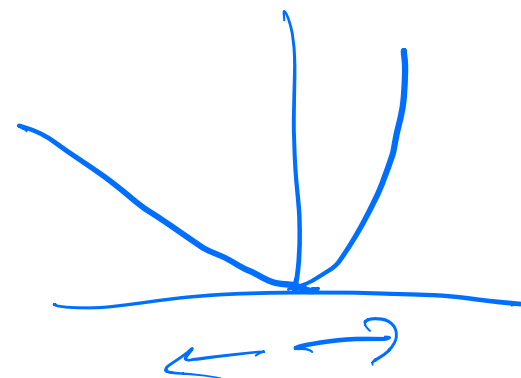
$$\mu(X) = E(Y | X = x)$$

This is our best possible choice of $\mu(X)$!

Regression function

$$|y - \hat{\mu}(x)|$$

$$\underbrace{\mu(X)}_{\text{true}} = \underbrace{\mathbb{E}(Y|X)}_{\text{goal}}$$



Given our setup with $(X, Y) \sim \mathcal{P}$ and assuming we really care about mean squared error loss, the regression function is the best we can possibly do.

Note: If we have a different loss, this ideal function *changes*.

However, we can't actually know $\mathbb{E}(Y|X)$, since we don't know $\mathcal{P}$! Instead, we try to get a good estimate of it from our data.

This gives us a lens for viewing many of the methods we will learn going forward.

Bias-variance decomposition

What happens when we try to estimate μ ? How can we be wrong?

To make notation simpler, write $\mu(x) = E(Y|X = x)$, our regression function. We are now estimating μ with $\hat{\mu}$ on a separate data set, so that is random too.

Then

$$\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X)) = E\left[E\left[(Y - \hat{\mu}(x))^2 | x\right]\right]$$

$$\begin{aligned} E\left[(Y - \hat{\mu}(x))^2 | x\right] &= E\left[(Y - \mu(x) + \mu(x) - \hat{\mu}(x))^2 | x\right] \\ &= E\left[(Y - \mu(x))^2 + (\mu(x) - \hat{\mu}(x))^2 + 2(Y - \mu(x))(\mu(x) - \hat{\mu}(x)) | x\right] \\ &= E\left[(Y - \mu(x))^2 | x\right] + E\left[(\mu(x) - \hat{\mu}(x))^2 | x\right] \\ &= E\left[(Y - E(Y|X))^2 | x\right] \end{aligned}$$

$$\text{Define } \sigma_x^2 = E\left[(Y - E(Y|X))^2 | x\right] = \text{Var}(Y|X)$$

Bias-variance decomposition

$$\begin{aligned}
 \mathbb{E}\mathcal{L}(Y, \hat{\mu}(X)|X=x) &= \sigma_x^2 + E[(\mu(x) - \hat{\mu}(x))^2|x] \\
 &= E[(\mu(x) - E(\hat{\mu}(x)|x) + E(\hat{\mu}(x)|x) - \hat{\mu}(x))^2|x] \\
 &= E[(\mu(x) - E(\hat{\mu}(x)|x))^2 + (E(\hat{\mu}(x)|x) - \hat{\mu}(x))^2 + 2(\mu(x) - E(\hat{\mu}(x)|x))(E(\hat{\mu}(x)|x) - \hat{\mu}(x)) |x] \\
 &= (\mu(x) - E(\hat{\mu}(x)|x))^2 + E[(\hat{\mu}(x) - E(\hat{\mu}(x)|x))^2|x]
 \end{aligned}$$

$$\text{bias}(\hat{\mu}(x)) =$$

$$\text{var}(\hat{\mu}(x)) =$$